TWEETING AWAY FIRM VALUE?

HOW INVESTORS EVALUATE CEOS' USE OF SOCIAL MEDIA

Hailiang Chen, Byoung-Hyoun Hwang, Baixiao Liu, and Yi Tang*

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This paper examines how investors react to CEOs' personal use of social media and how these reactions affect firm valuation. We theorize two mechanisms: a corporate communication channel, in which work-related tweets enhance transparency and increase firm value, and a CEO distraction channel, in which non-work-related tweets reduce value by signaling distraction or self-promotion. Using a panel of S&P 1500 firms from 2006–2020, covering 248 CEOs and over 250,000 tweets, we find that Tobin's Q rises with work-related tweeting volume but falls with non-work-related tweeting volume. These effects are moderated by information environment, CEO celebrity, and investor horizons. A supplementary survey of professional investors corroborates these mechanisms. Our results highlight social media's dual role as both a communication tool and an agency cost.

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* Chen and Tang are with the Faculty of Business and Economics, The University of Hong Kong, Pokfulam Road, Hong Kong. Liu is at HSBC Business School, Peking University, Xueyuan Boulevard, Shenzhen. Hwang is at Nanyang Business School, NTU, Nanyang Avenue, Singapore. All authors contributed equally. Emails: chen19@hku.hk, bh.hwang@ntu.edu.sg, baixiaoliu@phbs.pku.edu.cn, msytang@hku.hk. The authors would like to thank Robert Battalio, Tarun Chordia, Zhi Da, Theodore Goodman, Russell Jame, Yeejin Jang, Joon Mahn Lee, Tim Loughran, Yong Tan, Joseph Vithayathil, Jin Xu, Sean Xin Xu, Scott Yonker, and seminar participants at Boston College, City University of Hong Kong, Goethe University of Frankfurt, Korea University, Lingnan University, Nanyang Technological University, National University of Singapore, Seoul National University, Singapore Management University, University of Hong Kong, University of Zurich, York University, and the Workshop on Information Systems and Economics for their helpful comments. This research is supported by the Ministry of Education, Singapore, under its Academic Research Fund Tier 2 (022939-00001).

I. INTRODUCTION

Leaders, including heads of state, increasingly use personal social media accounts to engage with their constituents (PR Newswire, 2017).[2] Following this trend, a growing number of chief executive officers (CEOs) have adopted social media platforms such as Twitter to communicate with stakeholders.[3] Prominent examples include Tesla's Elon Musk and Salesforce's Marc Benioff, both of whom are habitual "tweeters."[4]

The increasing use of social media by CEOs raises important questions about how stakeholders react to this form of communication. A distinctive feature of CEOs' social media engagement is that it often extends beyond the workplace context (Heavey, Simsek, Kyprianou, and Risius, 2020), which complicates its interpretation. In this paper, we examine how investors, a critical stakeholder group, react to CEOs' personal social media activity and whether these reactions affect the valuations they assign to the firm.

There are aspects of social media engagement that investors likely view positively. CEOs possess comprehensive knowledge about their firms, and although (or, perhaps, because) social media communication is relatively informal, posts can in principle provide timely, unique and value-relevant insights to investors. CEOs' use of social media may also signal a commitment to transparency and openness, which investors should view favorably (Banerjee, Davis, and Gondhi, 2018). Recognizing that social media could improve a firm's overall information environment, the US Securities and Exchange Commission (SEC) has "*blessed the use of social media sites to broadcast market-moving corporate news*".[5]

But there are also potential drawbacks. As Drucker (1966) notes, "*Time is the scarcest resource,*

[2] *PR Newswire, "Twitter is the Prime Social Media Network for World Leaders"*
[3] In April 2023, Twitter was rebranded as "X." In this paper, we continue using the term "Twitter."
[4] *MIT Sloan Management Review, "How CEOs Can Leverage Twitter"*
[5] *WSJ, "SEC Embraces Social Media"*

and unless it is managed, nothing else can be managed." The time that CEOs devote to social media is time not spent on core responsibilities such as evaluating product and market trends, conducting strategic reviews, and cultivating relationships with current and future business partners. Because these tasks are critical to firm success, investors may interpret excessive social media activity as a sign of distraction and neglect of primary duties. Social media can also expose CEOs to attentional fragmentation (Leonardi and Meyer, 2015; Heavey et al., 2020; Risius, 2020), increase the risk of regulatory scrutiny,[6] and provoke consumer backlash.[7] Investors may view CEOs' social media activity despite these significant risks as flippant and self-serving rather than value-enhancing.

Drawing on these considerations, we theorize that investors evaluate CEOs' personal social media activity through two primary lenses. The first is a "corporate communication channel," in which work-related posts enhance transparency and improve firm valuation. The second is a "CEO distraction channel," in which non-work-related posts reduce valuation because they signal distraction or an undue focus on cultivating personal celebrity. When investors perceive social media activity as primarily serving communication purposes, they are likely to react positively. By contrast, when social media activity appears unrelated to corporate matters and suggests diversion of attention from core responsibilities, we expect investors to react negatively. We further propose three contextual factors that moderate the strength of these two channels: (1) the information environment, (2) CEO celebrity, and (3) investors' investment horizon. The first contextual factor primarily relates to the corporate communication channel, while the latter two are more pertinent to the CEO distraction channel.

Building on these theoretical arguments, we empirically examine CEOs of S&P 1500 firms

[6] https://www.ft.com/content/882520b8-8dfb-11e9-a24d-b42f641eca37
[7] https://blogs.vorecol.com/blog-the-role-of-social-media-in-shaping-corporate-reputation-9536

who, at some point during our 2006–2020 sample period, created an active personal Twitter account. We identify 248 such CEOs, who collectively posted 250,934 tweets. The evidence indicates that investors respond favorably to social media activity that provides value-relevant insights into company affairs, while they view self-oriented social media activity with concern. Specifically, panel regressions show that a firm's valuation ratio (Tobin's Q) increases with the volume of work-related tweets but decreases with non-work-related tweets. The positive association tied to work-related tweeting is stronger when firms face weaker analyst coverage and higher earnings forecast dispersion. By contrast, the negative association tied to non-work-related tweeting is amplified when non-work-related posts attract particularly high engagement and when firms have more long-term investors, who place greater emphasis on fundamentals and on governance structures that support sustainable value creation (McCahery, Sautner and Starks, 2016). Additional analyses reveal an inverted U-shaped relationship between the number of CEO tweets and firm valuation, suggesting that firm valuation rises with moderate CEO Twitter activity but declines when activity becomes excessive. Finally, we find no significant initial stock price reaction when CEOs first adopt Twitter, suggesting that investors do not respond to the mere adoption of social media but to the nature of its use, which only becomes observable over time.

To further validate our conclusions, we conduct a supplementary field survey in which we directly ask professional investors about their perceptions of CEOs' personal use of Twitter. The survey sample is unique in that it comprises investors from large U.S.-based financial institutions. The responses indicate that investors recognize both the potential benefits and drawbacks of CEO social media activity. They note its potential to enhance corporate communication, but they also stress that it can be a distraction and an agency cost. Overall, the survey results provide direct evidence that the two channels we conjecture to shape our econometric results feature prominently

in investors' own reasoning. These findings reinforce our regression-based evidence on how investors evaluate CEOs' social media activity.

Our paper contributes to the literature in several ways. First, we provide new insights into emerging research on CEOs' use of social media and its implications for firms (Heavey et al., 2020; Chen, Rui, and Whinston, 2021). Although interest in social media activity and its potential influence on firms has been growing, existing studies are either largely conceptual (Etter, Ravasi, and Colleoni, 2019; Heavey et al., 2020), focus on firms' corporate social media accounts (Castello, Etter, and Nielsen, 2016; Kim and Youm, 2017), or concerned with stakeholders' use of social media (Roccapriore and Pollock, 2023).

What remains underexplored is how important stakeholders view CEOs' personal social media activity and how such perceptions affect firms. Given CEOs' central role in shaping firm decisions and the growing prevalence of their social media engagement, this gap is striking. We help fill this gap by examining the inferences investors draw from CEOs' social media activity and the reasons such activity may be viewed positively or negatively.

Second, our study advances research on corporate communication and contributes to ongoing discussions about how modern information technologies affect firms. Prior studies on corporate communication have typically examined formal channels such as conference calls and official press releases (e.g., Pan, McNamara, Lee, Haleblian, and Devers, 2018; Shi, Zhang, and Hoskisson, 2019; Guo, Sengul, and Yu, 2021). By contrast, CEOs' use of social media represents a less formal but increasingly prominent channel for engaging key stakeholders (Heavey et al., 2020). Social media can be particularly effective for fostering connective and communal interactions with stakeholders (Kim and Youm, 2017; Etter et al., 2019). At the same time, the openness and informality of these platforms provide CEOs with wide latitude in how they use them. On one

hand, this flexibility could be what allows CEOs to disseminate unique, value-relevant information quickly, improving the information environment and increasing firm valuation. On the other hand, it can encourage self-centered, non-work-related activity, create distractions, and thereby destroy firm value. By documenting the balance between these opposing forces, our study informs the literature on corporate communication in the era of social media. Our findings also carry managerial and policy implications for executives, corporate boards, and financial regulators, who must weigh the informational benefits of CEOs' social media use against the potential distraction costs it imposes.

II. THEORY AND HYPOTHESES

A. CEOs' Use of Social Media

Social media has become pervasive in both personal and professional settings. While only 5 percent of adults in the United States reported using social media in 2005, this figure has risen to approximately 70 percent by 2019.[8] Firms, too, increasingly rely on social media as a medium to communicate with stakeholders. On the conceptual front, Etter et al. (2019) discuss how the rise of social media shapes the formation of corporate reputations. On the empirical front, Kim and Youm (2017) find that customers' reactions to firm-initiated social media messages significantly affect analyst stock recommendations. Castello et al. (2016), through an in-depth longitudinal case analysis of corporate stakeholder engagement, argue that corporations can develop legitimacy by participating in non-hierarchical open platforms and co-constructing agendas. More recently, Roccapriore and Pollock (2023) examine social media entrepreneurs and show that communication styles, along with the use of competence versus warmth cues, influence their ability to attract

[8] *Pew Research Center, "Social Media"*

followers and foster positive engagement through posts and comments.

All of the above studies focus on company-managed social media accounts. In today's corporate environment, CEOs increasingly use personal accounts to engage with stakeholders.[9] Heavey et al. (2020) propose a conceptual framework to organize, guide, and inspire research on CEOs' personal social media engagement; they develop a typology of CEOs' social media behaviors and advance propositions regarding the mechanisms, variation, and consequences of such engagement.

To the best of our knowledge, the only empirical evidence on CEOs' personal social media activity is provided by Chen et al.（2021), who find that CEO candidates use social media as a personal branding tool to improve their career prospects in the labor market. Beyond this study, we know relatively little about how key stakeholders, such as investors, evaluate CEOs' use of social media.

B. How Markets Respond to CEOs' Use of Social Media

In this subsection, we outline aspects of social media that investors may view positively as well as those they may view with concern. We focus on two mechanisms through which CEO tweeting can shape investor evaluations, though in opposite directions. We also consider contextual factors that moderate these effects. Overall, we argue that investors hold ambivalent views of CEOs' social media activity, with their reactions depending on the nature and content of the posts.

We begin with the potential benefits, which operate primarily through a "corporate communication channel." CEOs' social media posts often include firm- or product-related

[9] *Influential EXECUTIVE, "How Many Fortune 500 CEOs Are On Social Media In 2023?"*

announcements, as well as updates on their work-related day-to-day activities. Such content can provide investors with valuable insights into a firm's condition and prospects, thereby improving the information environment, especially when such information is less accessible through traditional, formal disclosure channels.

Before the advent of social media, direct communication from CEOs was rare. For instance, Kim and Meschke (2014) report that even among CEOs who appeared on CNBC, the average frequency was only 0.82 appearances per year. Similarly, direct communication from firms themselves can be limited. Although the SEC requires publicly traded firms to file Form 8-K to disclose significant events, firms exercise considerable discretion in determining what qualifies as significant. Consequently, there is wide variation in the number of 8-K filings across firms, even within the same industry. On average, firms file only eight Form 8-Ks annually, or roughly one every 46 days (Noh, So, and Weber, 2018). Media coverage can also be sparse. Even among large NYSE firms, more than one-quarter receive no media coverage in a given year; among NASDAQ firms, the corresponding fraction is almost 60 percent (Fang and Peress, 2009). Against this backdrop, investors may regard CEOs' social media posts as a valuable source of timely insights about their firms and as a meaningful enhancement to the overall information environment.

We argue that the potential concerns operate mostly through a "CEO distraction channel." Managerial attention is widely recognized as a scarce resource (March and Simon, 1958; Shepherd, McMullen, and Ocasio, 2017), and this is especially true for CEOs. Effective decision-making requires them to collect, process, and interpret information, activities that demand significant time and focus. To the extent that CEOs devote attention to social media for purposes unrelated to work, less attention remains available for tasks that are more critical to firm development. Shareholders may thus interpret CEO social media activity as a signal of distraction, negligence, and even a

form of agency cost.

Additionally, social media engagement exposes CEOs and their firms to significant downside risks. Financial regulators can view certain posts as misleading or in violation of Regulation Fair Disclosure. In fact, some CEOs' posts have already triggered investigations. For example, on September 26, 2024, the SEC charged DraftKings Inc. with selectively disclosing material, nonpublic information on the CEO's personal Twitter account.[10] Such investigations carry not only financial consequences but also further divert CEOs' attention away from core responsibilities. Social media posts can also provoke customer backlash. Notable examples include then-Twitter CEO Jack Dorsey encouraging travel to Myanmar,[11] then-Amazon CEO Jeff Bezos posting about dog sledding on Earth Day,[12] and former CrossFit CEO Greg Glassman's controversial tweet regarding George Floyd.[13] Given these risks, investors may interpret CEOs who are highly active on social media as flippant, self-serving, and inattentive to the best interests of the firm and its shareholders.

We propose that the content of a CEO's social media posts is the critical factor distinguishing the corporate communication channel from the CEO distraction channel. Specifically, we predict that work-related posts on social media enriches the firm's information environment and, in turn, enhances firm valuation. By contrast, when CEOs post non-work-related content, investors are more likely to view it as a distraction and respond less favorably, leading to lower valuations.

To expound on the former channel, if work-related posts indeed enrich the information environment, they can increase the value investors attach to the firm in two ways. First, an

[10] Other examples include *CNBC, "Netflix CEO's Facebook Post Triggered SEC Wells Notice"* and *NY Times, "SEC Broadens Inquiry Into Elon Musk's Disclosures About Twitter"*
[11] *Washington Post, "The latest public figure to face backlash for his tweet"*
[12] *Inc., "Amazon CEO Jeff Bezos Showed Off His Exciting Vacation on Twitter"*
[13] *CBS, "CrossFit CEO apologizes for offensive George Floyd tweet"*

enhanced information environment generally improves stock market liquidity and encourages broader investor participation, both of which enhance firm valuation (Diamond and Verrecchia, 1991). Second, greater transparency allows market valuations to more accurately reflect firms' intrinsic values (Lang, Lins, and Miller, 2003; Fernandes and Ferreira, 2008). Because a substantial portion of CEO pay is equity-based, more accurate stock prices strengthen incentive alignment between managers and shareholders (Jensen and Meckling, 1976), thereby lowering agency costs and further increasing firm value. These arguments form the basis of the first part of our first hypothesis:[14]

Hypothesis 1a ("H1a"): There is a positive relationship between CEOs' work-related social media activity and firm valuation.

By contrast, when CEOs primarily post non-work-related content, they are more likely to use social media for self-centered, personal purposes including social recognition and psychological comfort (Jan, Soomro, and Ahmad, 2017). Investors likely view such behavior as a misallocation of a CEO's scarce attention and as evidence of weak alignment with shareholder interests.

Non-work-related posts are also more prone to sparking controversy or consumer backlash (Pantano, 2021), since they more frequently touch on sensitive or divisive topics that could alienate stakeholders. Such posts can inadvertently damage the company's reputation and generate public relations challenges that further divert resources from core operations (Heavey et al., 2020). In addition, non-work-related content increases the risk of regulatory breaches or the dissemination of misleading information (Wang, Reger, and Pfarrer, 2021). Costly investigations and legal scrutiny can ensue, exacerbating concerns about managerial judgment and discipline. A CEO's

[14] We note that it is by no means certain that investors will view CEOs' social media posts as enhancing the information environment and attach higher valuations to the corresponding firms. If the relationship in Hypothesis 1a were entirely straightforward, there would be little need for empirical evaluation. What matters in the development of Hypothesis 1a (and all our subsequent hypotheses) is that the hypothesis is theoretically sound and defensible.

willingness to engage in non-work-related posting despite these risks can generate the impression that they are flippant and self-serving, further raising agency costs. The second part of our first hypothesis is thus:

Hypothesis 1b ("H1b"): There is a negative relationship between CEOs' non-work-related social media activity and firm valuation.

C. Boundary Conditions

Because CEOs' social media activity can significantly affect firm valuation, it is important to identify the conditions under which this impact is either amplified or mitigated. We therefore identify factors that shape whether investors interpret a CEO's social media engagement primarily through the lens of the corporate communication channel or the CEO distraction channel. When circumstances lead investors to view social media activity primarily through the corporate communication lens, the positive effects on firm valuation should be strengthened. Conversely, when circumstances cause investors to view such activity primarily as a distraction from core responsibilities, the negative consequences for valuation should be magnified.

We focus on three layers of boundary conditions. At the firm level, we analyze the information environment. At the CEO level, we investigate the role of CEO celebrity. At the investor level, we examine their investment horizons.

The first moderating factor is the firm's information environment. Two salient indicators for the information environment are the level of sell-side analyst coverage and the degree of analysts' earnings forecast dispersion. Sell-side analysts play a central role in financial markets by gathering and analyzing firm-specific information and disseminating their insights to investors. When analyst coverage is extensive, the firm's information environment is relatively rich. By contrast,

when coverage is limited, less information is produced, leaving investors with greater information gaps. In such contexts, CEO social media activity may serve as an important complementary source of information. Work-related posts allow CEOs to communicate directly with stakeholders, thereby improving investors' understanding of firm operations and strategy (Zavyalova, Pfarrer, Reger, and Shapiro, 2012). Accordingly, we conjecture that when analyst coverage is low (and thus the need for additional information is high), investors are more likely to interpret CEOs' social media activity through a corporate communication channel and respond more favorably.

Hypothesis 2a ("H2a"): The lower the analyst coverage of a firm, the stronger the positive relationship between a CEO's work-related social media activity and firm valuation.

Relatedly, when analysts exhibit strong agreement about a firm's future prospects, investors operate in an information environment with relatively little ambiguity. By contrast, when analyst forecasts diverge, investors face heightened uncertainty about the firm's outlook. In such contexts, CEOs' work-related social media posts can play a clarifying role. CEOs can use social media as a vehicle to address market concerns and explain strategic decisions to investors. They may also communicate the firm's culture, values, and long-term vision, which enables investors to develop a more comprehensive understanding of the company and assess its trajectory more effectively (Westphal and Graebner, 2010; Graffin, Carpenter, and Boivie, 2011). We therefore predict that when analyst earnings forecast dispersion is high (and thus the need for clarifying information is high), investors are more likely to interpret CEOs' social media activity through a corporate communication channel and respond more favorably.

Hypothesis 2b ("H2b"): The higher the analyst earnings forecast dispersion of a firm, the stronger the positive relationship between a CEO's work-related social media activity and firm valuation.

The second moderating factor we consider is CEO celebrity, which we expect to operate primarily through the CEO distraction channel. Some CEOs achieve high levels of engagement on social media, where their posts attract substantial numbers of likes and retweets. Although such popularity may generate favorable reactions from certain stakeholders, such as consumers, it can also carry adverse implications for firm valuation, particularly when the most visible posts are non-work-related.

When a CEO becomes a social media celebrity for their non-work-related posts, they may increasingly enjoy the virtual limelight and devote less attention to core business responsibilities. Prior research indicates that celebrity status does not improve decision-making. Hayward, Rindova, and Pollock (2004) argue that CEO popularity can heighten overconfidence and hinder critical strategic changes. Wade, Porac, Pollock, and Graffin (2006) find that firms with more popular CEOs often perform worse even as those CEOs receive higher compensation. Malmendier and Tate (2009) similarly demonstrate that "superstar CEOs" (e.g., award winners) tend to underperform. One reason, the authors argue, is that they devote substantial time to activities outside their core responsibilities, such as writing autobiographies.

Building on this evidence, we propose that when investors observe CEOs sending non-work-related posts that attract very high levels of user engagement, investors are more likely to fear distraction and undue focus on cultivating personal celebrity and, as a result, react negatively.

Hypothesis 3 ("H3"): The more popular a CEO is on social media, the stronger the negative relationship between non-work-related social media activity and firm valuation.

The third moderating factor pertains to investors' investment horizons. Investors differ in their holding periods, which influence whether they emphasize short- or long-term outcomes (Bushee, 1998; Connelly, Hoskisson, Tihanyi, and Certo, 2010; Brochet, Loumioti, and Serafeim, 2014).

Some investors maintain short horizons and trade frequently (e.g., "transient" investors in Bushee, 1998). Because of their frequent turnover, these investors place greater weight on short-term performance fluctuations, regardless of whether those fluctuations are tied to fundamentals (Cremers, Pareek, and Sautner, 2020).

By contrast, in the long run, market prices should converge to fundamentals (Fama, 1970). Long-term oriented investors should therefore place greater emphasis on fundamentals and on governance structures that support sustainable value creation (Chen, Harford, and Li, 2007; McCahery et al., 2016). For such investors, observing a CEO engage in non-work-related social media activity heightens concerns about managerial distraction from tasks that are essential to long-term performance.

We therefore predict that when firms have a larger proportion of long-term investors, the negative effect of CEOs' non-work-related social media activity on firm valuation will be more pronounced.

Hypothesis 4 ("H4"): The higher percentage of long-term investors in a firm, the stronger the negative relationship between a CEO's non-work-related social media activity and firm valuation.

III. METHOD

A. Sample

We test our predictions using a sample of CEOs of publicly traded firms in the US. First, we compile a list of all CEOs in Execucomp from 2006 through 2020. Execucomp covers the top executives of S&P 1500 companies, including those that were previously part of the index but are still trading. We begin in 2006, coinciding with the founding of Twitter, and end in 2020, as our data collection commenced in early 2021. Our list of all CEOs in Execucomp comprises 5,242

CEOs working for 2,738 distinct companies.

To identify whether a CEO has a personal Twitter account, we recruited six research assistants and provided them with the list of the 5,242 CEOs. The list contained each CEO's first name, last name, and company affiliation. We tasked the research assistants to check for each CEO whether they had an active personal Twitter account. Each CEO was searched by two research assistants. When they found a potential match, we cross-referenced the Twitter account with personal information from Execucomp, including the CEO's age, gender, and past work experience. We also reviewed the Twitter profile and examined several tweets. To ensure accuracy, we took a conservative approach and excluded accounts that we could not confidently verify as belonging to the CEO in question. Ultimately, we determined that 248 CEOs have authentic, active personal Twitter accounts. For each of these accounts, we collect the account identifier ("screen name"), personal biography, account registration date, and number of followers as of May 2021.

We use two approaches to download the CEOs' tweets, capturing both original tweets and retweets/replies to other posts. First, we write a program using Twitter's official API (https://developer.twitter.com) to retrieve all timeline tweets for each CEO's account. The official API only provides access to the most recent 3,200 tweets. To overcome this limitation, we supplement the API approach with the "snscrape" Python package, a popular tool for scraping data from social media. This second approach allows us to collect older tweets via Twitter's advanced search function (https://twitter.com/search-advanced). In total, we retrieve 250,934 tweets posted by 248 CEOs from 2008 to 2020. For each tweet, we collect the account identifier ("screen name"), tweet identifier, date, time, content, number of likes, and number of retweets.

We categorize each tweet in our sample as work-related or non-work-related. Given the large volume of tweets, we use a combination of manual labeling and machine learning. First, we

recruited eight research assistants to manually categorize a subset of 34,585 tweets into three categories: "Type-1 tweets," which include company-related news announcements; "Type-2 tweets," which describe a CEO's day-to-day work activities; and "Type-3 tweets," which reflect a CEO's personal interests, hobbies, or other non-work-related activities. Each tweet was evaluated by three research assistants, with the majority vote determining the label. If there is no majority, we have a fourth assistant break the tie. In our analysis, we consider Type-1 and Type-2 tweets as work-related tweets and Type-3 tweets as non-work- related tweets.[15]

An example of a Type-1 tweet is: "*Tesla has suspended vehicle purchases using Bitcoin. We are concerned about rapidly increasing use of fossil fuels for Bitcoin mining and transactions, especially coal, which has the worst emissions of any fuel. [...]*" (tweet by Elon Musk, CEO of Tesla, 5/13/2021).

Examples of Type-2 tweets include: "*Earnings call. T-1 hr away. I enjoy taking a step back from the day to day and reflecting on all we have accomplished over the past qtr*" (tweet by John Heyman, CEO of Radiant Systems, 10/29/2009), and "*Meeting with our partner and customer in Shanghai; Mr. Chen Hong, President of SAIC. Helping me eat. http://t.co/Jik93eMMxZ*" (tweet by Alex Molinaroli, CEO of Johnson Controls, 2/26/2014).

Examples of Type-3 tweets include: "*Dinner at Hammersley's in Boston—this is still a great restaurant!!*" (tweet by George F. Colony, CEO of Forrester Research, 10/23/2008), and "*Heading to the @Aaarena for the BIG @MiamiHEAT OKC Thunder match-up. Tip is 8pm sharp be there loud & in Black.*" (tweet by Micky Arison, CEO of Carnival, 4/4/2012).

To categorize the remaining 216,349 tweets, we build a machine-learning model based on the manually-labeled data. We randomly split the human-coded tweets into a training set and a testing

[15] We group Type 1 and Type 2 into a single "work-related" category in the main analysis as Type 1 tweets end up accounting for only 0.06% of the sample and offer limited variation.

set in a 70:30 ratio. We use the training set to train the model and evaluate its performance using the testing set. We consider five popular classification algorithms: Logistic Regression (LR), Naïve Bayes (NB), Random Forest (RF), Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). The SVM algorithm performs the best, so we use it to categorize the remaining tweets. Results from the other algorithms are similar and available upon request. We provide more information on our machine learning method in Online Appendix A. Our final sample covers 2,551 S&P 1500 firms and 248 CEOs who tweet at some point during our sample period. As of 2020, firms with tweeting CEOs make up 33% of the total market capitalization of the S&P 1500.

B. Variables

Dependent Variable

We use Tobin's Q (*TBQ*) to measure the value that investors assign to a specific company in a particular year (Dezső and Ross, 2012; Jayachandran, Kalaignanam and Eilert, 2013; Awaysheh, Heron, Perry and Wilson, 2020). *TBQ* is defined as a firm's market value of assets, divided by the book value of its assets. It is calculated in COMPUSTAT as:

$$TBQ = \frac{AT - CEQ + PRCC \times CSHO}{AT}$$

where *AT* represents total assets, *CEQ* is shareholders' equity, *PRCC* is the closing price of common stock, and *CSHO* is the number of shares outstanding.

Key Independent Variable

To examine how investors respond to CEOs' Twitter activity, we conduct a panel regression analysis using the full Execucomp universe from 2008 to 2020. The unit of observation is at the firm-year level. Our key independent variables are *Ln(#Work-Related Tweets)* and *Ln(#Non-Work-*

Related Tweets), defined as the natural logarithm of one plus the number of work-related and non-work-related tweets, respectively, that the CEO posts in the corresponding year. We also construct *Ln(#Tweets)*, the natural logarithm of one plus the CEO's total number of tweets, along with its squared term, to test for a possible curvilinear relationship between the volume of CEO tweets and firm valuation.

To test Hypotheses 2 through 4, we conduct heterogeneity analyses by interacting the variables *Ln(#Work-Related Tweets)* and *Ln(#Non-Work-Related Tweets)* with (1) measures of the firm's information environment, (2) measures of the CEO's celebrity status, and (3) measures of the firm's investor base.

We use *Inverse Analyst Coverage* to capture the need for CEO communication. This variable is defined as the annual average of one divided by one plus the total number of sell-side analysts issuing quarterly earnings-per-share forecasts for the corresponding firm in the respective year. We also construct *Analyst Dispersion*, measured as the annual average of the standard deviations of sell-side analysts' quarterly earnings-per-share forecasts, scaled by the firm's stock price as of the corresponding fiscal quarter end. High realizations of *Inverse Analyst Coverage* and *Analyst Dispersion* imply a greater need for additional, clarifying information.

To assess a CEO's celebrity status from his non-work-related Twitter activity, we create *Ln(#Likes-Non-Work-Related)* and *Ln(Retweets-Non-Work-Related)*, defined as the natural logarithm of one plus the number of likes and retweets, respectively, that the CEO receives on their non-work-related tweets in the corresponding year.

To measure the composition of a firm's investor base, we use *%Long-Term*, which captures the fraction of shares held by "long-term investors" in the corresponding year. Following Bushee (1998), long-term investors are defined as institutional investors that are not "transient."

Specifically, Bushee calculates portfolio turnover rates from publicly disclosed quarterly institutional investors holdings data to categorize investors as transient, dedicated, or quasi-indexers. Transient investors are characterized by the shortest investment holding periods.

Control Variables

In our panel regressions, we include a set of CEO- and firm-level control variables. We detail the construction of these variables in Online Appendix B. Our CEO-level control variables include *CEO Age, CEO Tenure, Male CEO, Ln(Total Compensation), Extravert.*

Our firm-level control variables include *Inst. Holdings, Size, Cashflow, ROA, Leverage, Leverage, Dividend, Capital Expenditure, R&D, Sales/Total Assets, Sales Growth, Stock Returns, Loss, Z-Score,* and *E-Index*. The *Z-Score* predicts the likelihood of bankruptcy based on various financial ratios, such as the ratio of working capital to total assets (Altman, 1968). The *E-Index* measures the quality of a firm's corporate governance based on six entrenchment indicators, such as having a staggered board or poison pills (Bebchuk, Cohen, and Ferrell, 2009).

We also account for the firm's information environment by including *Ln(#News Articles)* from the Dow Jones Newswires, *Inverse Analyst Coverage,* and *Analyst Dispersion*.

IV. EMPIRICAL DESIGN

A. Full Panel Regression Model

To assess how investors evaluate the extent of a CEO's Twitter usage, we first examine whether firm valuation correlates with the number of work-related and non-work-related tweets by the CEO.

Our panel regression model is specified in regression equation (1):

$$TBQ_{i,t} = \alpha_i + \delta_t + \beta_1 Ln(\#Work - Related\ Tweets)_{i,t} + \beta_2 Ln(\#Non - Work - Related\ Tweets)_{i,t} + X_{i,t}\beta \quad + \epsilon_{i,t}\ , \tag{1}$$

where the unit of analysis is firm-year. The dependent variable, $TBQ_{i,t}$, represents firm i's Tobin's Q in year t. α_i are firm fixed effects, which control for time-invariant firm characteristics. δ_t are year fixed effects, which account for common shocks affecting all firms.

Our primary variables of interest are $Ln(\#Work\text{-}Related\ Tweets)_{i,t}$ and $Ln(\#Non\text{-}Work\text{-}Related\ Tweets)_{i,t}$. The coefficient estimates on these variables capture the partial correlations between a CEO's work-related and non-work-related Twitter activity and firm valuation. $X_{i,t}$ represents the set of control variables. We compute robust standard errors.

We also test whether CEO Twitter activity exhibits a curvilinear relationship with firm valuation. Our panel regression model is specified in regression equation (2):

$$TBQ_{i,t} = \alpha_i + \delta_t + \beta_1 Ln(\#Tweets)_{i,t} + \beta_2 Ln(\#Tweets)_{i,t}\text{^}2 + X_{i,t}\beta \quad + \epsilon_{i,t} \tag{2}$$

This specification mirrors equation (1), except the independent variables of interest are $Ln(\#Tweets)_{i,t}$ and its squared term. The coefficient estimates on these variables indicate whether firm valuation rises with moderate CEO Twitter activity but declines when activity becomes excessive.

B. Short-Run Event Study

To assess how investors react to CEOs' initial adoption of Twitter, we examine the immediate stock price responses around the dates when CEOs send their first tweets. Short-term market reactions provide insight into investors' initial perceptions of CEO engagement with social media. To control for confounding events, we also separately examine investors' reaction to CEOs' initial

adoption of Twitter where on the date of the adoption, there is no news articles published about the CEO's firm in the Dow Jones Newswires.

We define the date a CEO posted their first tweet as the CEO's "Twitter adoption date." We measure stock price responses using abnormal returns, defined as the firm's daily stock return minus the corresponding CRSP value-weighted market return. We then compute cumulative abnormal returns (*CARs*) over three event windows: (0, +1), (0, +2), and (0, +3), where day 0 represents the CEO's Twitter adoption date. We compute robust standard errors.

V. RESULTS

A. Descriptive Statistics

In Table 1, we present summary statistics on CEOs' tweeting activities. Across our sample of 248 CEOs, the average CEO posts 1,012 tweets over the sample period, or approximately 119 tweets per year. On average, these tweets receive 115,494 likes and 55,452 retweets annually. The distribution of tweeting activity is highly right-skewed: the top ten tweeting CEOs post more than 600 tweets per year. The level of attention that CEOs' posts receive is also heavily skewed to the right. For example, Elon Musk, the CEO of Tesla, though not the most active in terms of tweeting volume, received on average more than 23 million likes and 2.5 million retweets per year.

[Insert Table 1 here]

In Table 2, we report the means and standard deviations for all the variables used in our panel regression model.

[Insert Table 2 here]

B. CEO Twitter Usage and Firm Valuation

Table 3 reports the regression results from our full panel analysis. In Models 1 and 2, we evaluate the effects of CEOs' work- and non-work-related Twitter volume on firm valuation. The results show that Tobin's Q is significantly higher when CEOs post more work-related tweets (β_1 = 0.10, *p*-value = 0.01) and lower when they post more non-work-related tweets (β_2 = -0.09, *p*-value = 0.06). To put these estimates in perspective, the mean Tobin's Q in our sample is 1.87. A one-standard-deviation increase in work-related tweeting is associated with a 3.5 percent increase in Tobin's Q, whereas a one-standard-deviation increase in non-work-related tweeting is associated with a 3.0 percent decline. These partial correlations are consistent with Hypotheses H1a and H1b: investors generally view CEOs' work-related social media activity positively and non-work-related activity negatively.

[Insert Table 3 here]

Model 3 further shows that CEO Twitter volume exhibits a curvilinear relationship with Tobin's Q ((β_1 = 0.17, *p*-value = 0.00 and β_2 = -0.03, *p*-value = 0.00). At lower levels of activity, tweeting volume positively associates with valuation. However, beyond a threshold of roughly 213 tweets per year, increased CEOs' social media activity negatively relates to firm valuation.

[Insert Figure 1 here]

Figure 1 plots this curvilinear pattern alongside the share of work-related tweets. For CEOs in the bottom decile of annual tweeting volume, 64.8 percent of their posts are work-related. However, this fraction declines steadily and falls to 41.6 percent among CEOs in the top decile of annual tweeting volume. Thus, the curvilinear relationship between Tobin's Q and the number of tweets appears to at least partially stem from a shift in content: initial tweets focus on value-relevant information, which investors appreciate, but as volume increases, work-related content becomes diluted by non-work-related posts, which investors interpret less favorably.

In Table 4, we present the results from testing the moderating effects proposed in Hypotheses H2 through H4. Columns 1 and 2 report the results for H2a and H2b, which concern the role of the information environment. As shown in Column 1, firms with higher *Inverse Analyst Coverage* exhibit a more positive correlation between valuation and volume of work-related tweeting (β= 0.07, *p*-value = 0.15). The positive correlation is consistent with H2a, but we note that the statistical significance is low. Similarly, Column 2 shows that the positive correlation is stronger for firms with higher *Analyst Dispersion* (β= 0.03, *p*-value = 0.01). Taken together, these findings support H2a and H2b, suggesting that when investors face weaker or more ambiguous information environment, they place greater value on direct communication from CEOs, amplifying the positive valuation effect of work-related social media activity.

[Insert Table 4 here]

Panel A of Online Appendix C plots the marginal effects of CEOs' work-related tweeting activity on Tobin's Q across varying levels of *Inverse Analyst Coverage* and *Analyst Dispersion.* The figure shows that the positive association between the volume of work-related tweeting and

firm valuation strengthens as either moderator increases. This visual evidence is consistent with H2a and H2b.

Columns 3 and 4 present the results for H3, which concerns the impact of CEOs' celebrity status. The interaction terms indicate that firms whose CEOs receive more likes and retweets on their non-work-related tweets exhibit a more pronounced negative correlation between non-work-related tweeting volume and firm valuation (β = -0.09 and -0.02, *p*-values = 0.00 and 0.00, respectively). These results suggest that when CEOs attract greater user engagement on their non-work-related posts, the adverse impact of such activity on firm valuation intensifies.

Column 5 indicates that firms with a higher proportion of long-term investors experience a more pronounced negative valuation effect of non-work-related tweeting activity (β = -0.06, *p*-value = 0.08). These findings support H4 that long-term investors are particularly concerned about CEO distraction from non-work-related social media activity.

Panel B of Online Appendix C plots the marginal effects of CEOs' non-work-related tweeting on Tobin's Q across varying levels of the moderators: the number of likes received for non-work-related tweets, the number of retweets received for such tweets, and the proportion of long-term investors. The figure shows that the negative association between non-work-related tweeting volume and firm valuation becomes stronger when these moderators are high, consistent with H3 and H4.

C. Robustness Checks

We conduct several additional analyses to assess the robustness of our baseline results. First, we exclude Tesla from the sample to mitigate concerns that Elon Musk may unduly influence the findings. Second, we double-cluster standard errors at both the firm and year levels to account

for serial correlation and heteroskedasticity. Third, we lag all independent variables by one year to address potential reverse causality concerns.[16] Finally, we winsorize all variables at the 1st and 99th percentiles to reduce the impact of outliers. The results, reported in Online Appendix D, remain economically meaningful, although the coefficient estimates in some specifications are only marginally statistically significant.

D. CEO Twitter Adoption and Stock Returns

Table 5 presents the results from our event study. Columns 1 through 3 report the market-adjusted cumulative abnormal returns (*CARs*) for the full sample of 248 CEO Twitter adoptions over two-day, three-day, and four-day windows. The CARs are 0.24%, 0.33%, and 0.39%, with *p*-values of 0.33, 0.33, and 0.32, respectively. Columns 4 through 6 report the CARs for the subsample of 194 CEO Twitter adoptions that are not accompanied by other news coverage of the CEO's firm. The corresponding CARs are 0.22%, 0.37%, and 0.42%, with *p*-values of 0.41, 0.39, and 0.39, respectively.

Overall, these findings suggest that in the short run, investors react ambivalently to CEO Twitter adoptions. On average, the stock market response is neither significantly positive nor significantly negative. At the time of adoption, investors may remain uncertain about whether a CEO's Twitter activity will ultimately benefit or harm the firm.

[Insert Table 5 here]

[16] While our strategies to gauge the relevance of endogeneity concerns (using lagged independent variables, controlling for prior stock performance, and conducting moderating analyses) reduce these concerns to some extent, they cannot fully eliminate them. Our results should therefore be interpreted with appropriate caution.

E. Survey-Based Evidence

Our regression-based analyses infer investors' reactions to CEOs' social media activity from observed correlations between firm valuations and the level of Twitter activity. While these findings are consistent with our hypotheses, two concerns remain. First, even with firm- and year-fixed effects and our extensive set of control variables, we cannot rule out that unobserved events, which happen to coincide with the volume of tweeting, drive the results. Second, although our patterns are consistent with corporate communication and CEO distraction, we have not yet provided direct evidence that investors explicitly consider these factors.

To address these concerns, we complement our econometric analysis with direct survey evidence. We ask investors whether they view CEOs' Twitter activity positively or negatively, and which considerations matter most in shaping their views.

Because institutional investors dominate both ownership and trading in U.S. equity markets, holding about 75% of the shares of our sample firms and accounting for over 85% of trading volume during our sample period (Bloomberg Intelligence, 2021), we tried very hard to target this group in particular. Partnering with CoreData Research, we surveyed 100 U.S.-based institutional investors in November of 2022. As shown in Online Appendix Table E-1, the respondents are highly experienced: 81% report managing assets exceeding $100 million, 34% report managing more than $2.5 billion, and 97% have over ten years of experience. We used carefully worded open-ended questions to minimize framing bias.

The results, reported in Online Appendix E, reveal that investors perceive two main benefits of CEO Twitter activity: (1) valuable information that improves the firm's information environment and (2) enhanced visibility to customers and investors. At the same time, investors cite three main drawbacks: (1) legal and regulatory risks, including potential charges of market

manipulation; (2) the danger of alienating customers or investors through controversial posts; and (3) the perception that social media engagement reflects flippant or self-serving leadership misaligned with shareholder interests. Many respondents emphasize that personal use of Twitter exposes firms to reputational, legal, and financial risks, and that a CEO's willingness to accept these risks signals poor judgment.

Taken together, these findings corroborate our central hypotheses. Investors confirm that they value CEOs' social media activity when it enhances corporate communication (H1a), but they also stress that it can be a distraction and agency cost (H1b). In this way, the survey responses provide direct evidence that the same two channels conjectured to shape our econometric results also feature prominently in investors' own reasoning.

VI. DISCUSSION

The central message of our study is that CEOs' use of social media carries both benefits and costs for their firms. Our analyses indicate that investors interpret work-related social media activity as a valuable channel of corporate communication, one that improves the information environment and enhances firm valuation. By contrast, they view non-work-related activity as a form of agency cost that distracts managerial attention from core responsibilities. Consequently, firms whose CEOs engage more in work-related tweeting tend to trade at higher valuations, whereas firms whose CEOs focus more on non-work-related tweeting tend to trade at lower valuations.

A. Implications

Our findings carry important implications for research on corporate communication and

governance. Social media has become a pervasive part of modern life, and many prominent CEOs, such as Elon Musk, actively engage on these platforms. It is therefore natural to ask whether and how such activity influences their firms. Existing management research has largely remained silent on this issue. By addressing this gap, our study provides novel evidence on how investors evaluate an increasingly common "off-the-job activity" of CEOs and the implications of such activity for firm valuation.

Our findings also align with the view that the consequences of executive actions depend critically on contextual conditions (e.g., Busenbark, Krause, Boivie, and Graffin, 2016; Neely, Lovelace, Cowen, and Hiller, 2020). As Neely et al. (2020) note, moderating factors have been a central focus of executive research over the past decade. We emphasize that there is no universal answer to whether CEOs' social media engagement is beneficial or detrimental. Rather, the effects hinge on the content of their activity and are further shaped by the firm's information environment, the extent of CEO celebrity, and the investment horizons of the shareholder base.

From a practical perspective, our results highlight the importance of aligning executive behavior with shareholder interests through effective governance mechanisms. While corporate governance research has traditionally concentrated on executives' on-the-job decisions, our findings suggest that certain off-the-job activities, such as personal social media use, could also be subject to governance oversight (Ouyang, Tang, Wang, and Zhou, 2022).

B. Limitations and Future Research

Our study has several limitations, which in turn open avenues for future research. First, our measure of CEOs' social media activity is limited, as we focus solely on Twitter accounts. CEOs engage on other platforms that differ in audience, style, and influence. Excluding these platforms

likely biases our results downward by reducing the variance in our key independent variable, making it more difficult to find support for our hypotheses (Hambrick and Mason, 1984; Ouyang et al., 2022). Nevertheless, developing a broader measure that captures multi-platform activity would be valuable for validating and extending our conclusions.

Relatedly, future research could benefit from methodological extensions that complement our design. One promising avenue is the application of advanced natural language processing tools to analyze not only the volume and broad category of posts but also the nuanced topics and framing across time. Moreover, incorporating cross-platform data could provide a more holistic view of CEOs' digital presence and the consistency (or inconsistency) of their messaging. Such methodological innovations would allow scholars to capture a richer and more nuanced understanding of how CEO social media behavior shapes investor perceptions and firm outcomes.

Third, social media engagement could shape a wide range of corporate decisions and outcomes (Hambrick and Mason, 1984; Heavey et al., 2020). Future research could investigate how a CEO's social media behavior relates to outcomes such as mergers and acquisitions, strategic alliances, or innovation. For example, if habitual social media activity reflects narcissistic tendencies, one might expect stronger associations with acquisition intensity or empire-building initiatives.

Fourth, our findings reflect broad, average effects. But there are prominent outliers such as Elon Musk and Tesla. Elon Musk frequently engages in non-work-related posting, yet under his leadership Tesla's valuation has risen to extraordinary levels over our sample period. Such "corner situations" suggest that non-work-related tweeting can, under certain conditions, cultivate a devoted investor base and sustain narrative-driven valuations over extended horizons. Identifying the boundary conditions under which CEO non-work-related social media activity enhances rather than erodes long-term market valuations should be an interesting avenue for future research.

Finally, our analysis focuses on publicly traded firms in the United States. Our findings may not generalize to private firms or to firms operating in other institutional or cultural settings, where managerial discretion, governance structures, and stakeholder expectations differ. In some cultures, investors may perceive CEO social media activity as particularly unprofessional, while in others investors may hold the view that the actual time that CEOs spend on posts must be limited and could not materially interfere with managerial responsibilities. Future work could examine whether our conclusions vary across different ownership structures and institutional and cultural contexts (Wiersema and Bird, 1996; Crossland and Hambrick, 2011).

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Figure 1: CEO Twitter Activities, Firm Tobin's Q, and % CEO Work-Related Tweets

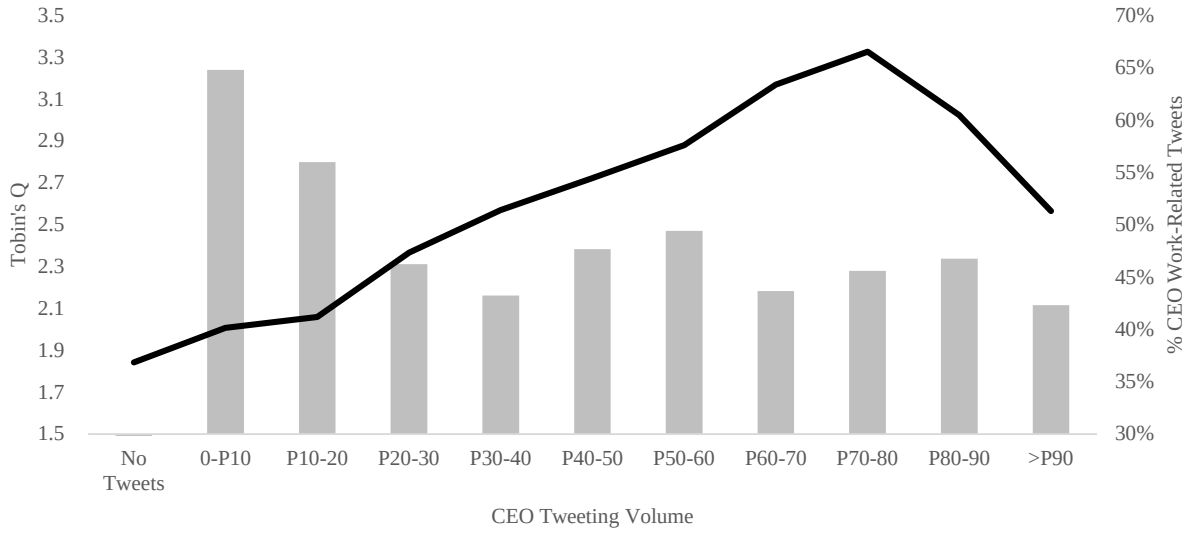


Table 1: Tweeting CEO Statistics

	Twitter Adoption Date	Total #Tweets	#Tweets/ Year	#Likes/ Year	#Retweets/ Year
All Tweeting CEOs (248)		1,012	119	115,495	55,452
Top 10 Tweeting CEOs					
Fernando Aguirre (Chiquita Brands Intl Inc)	2011/3/3	21,958	1,996	1,614	18,318
Douglas Conant (Campbell Soup Co)	2011/5/16	16,661	1,515	10,401	6,536
Michel Combes (Sprint Corp)	2018/6/1	4,182	1,394	29,199	22,119
Micky Arison (Carnival Corporation & Plc)	2012/3/30	9,961	1,107	22,831	130,360
Elon Musk (Tesla Inc)	2010/6/4	12,826	1,069	23,866,070	2,518,766
Marc Benioff (Salesforce.Com Inc)	2009/2/28	13,366	1,028	140,896	325,402
Wallace Boston (American Public Education)	2009/3/12	11,684	899	263	3,606
Scott Ginsburg (Sizmek Inc)	2009/7/4	4,653	665	216	17,686
David Barger (JetBlue Airways Corp)	2011/1/19	1,957	652	155	13,306
Patrick Decker (Xylem Inc)	2014/8/1	5,150	644	1,764	815,343

Table 2: Descriptive Statistics

Variables	Mean	SD
Dependent and Key Independent Variables		
Tobin's Q	1.87	1.64
Ln(#Work-Related Tweets)	0.11	0.66
Ln(#Non-Work-Related Tweets)	0.10	0.63
Ln(#Tweets)	0.13	0.76
Moderators		
Inverse Analyst Coverage	0.28	0.33
Analyst Dispersion	0.10	3.58
Ln(#Likes-Non-Work-Related)	0.12	0.86
Ln(#Retweets-Non-Work-Related)	0.16	1.11
%Long-Term	0.35	0.61
Control Variables		
CEO Age	56.46	7.28
CEO Tenure	8.29	7.34
Male CEO	0.96	0.20
Ln(Total Compensation)	8.20	1.13
Extravert	0.23	0.57
Inst. Holdings	0.78	0.25
Size	0.40	0.26
Cash Flow	0.08	0.13
ROA	0.04	0.13
Leverage	0.25	0.22
Dividend	0.16	0.83
Capital Expenditures	0.04	0.08
R&D	0.03	0.07
Sales/Total Assets	0.88	0.89
Sales Growth	0.07	0.21
Stock Returns	0.14	0.56
Loss	0.11	0.12
Z-Score	1.17	4.34
E-Index	3.04	1.08
Ln(#News Articles)	2.43	2.10

Table 3: CEO Twitter Activities and Firm Performance

	(1) Tobin's Q	(2) Tobin's Q	(3) Tobin's Q
Ln(#Work-Related Tweets)	0.10	0.10	
	(0.03)	(0.01)	
Ln(#Non-Work-Related Tweets)	-0.09	-0.09	
	(0.08)	(0.06)	
Ln(#Tweets)			0.17
			(0.00)
Ln(#Tweets)^2			-0.03
			(0.00)
CEO Age		-0.00	-0.00
		(0.72)	(0.68)
CEO Tenure		0.00	0.00
		(0.03)	(0.03)
Male CEO		-0.02	-0.02
		(0.60)	(0.62)
Ln(Total Compensation)		0.05	0.05
		(0.00)	(0.00)
Extravert		-0.03	-0.03
		(0.04)	(0.03)
Inst. Holdings		-0.08	-0.08
		(0.04)	(0.04)
Size		-0.14	-0.12
		(0.45)	(0.51)
Cash Flow		-0.00	-0.00
		(0.08)	(0.08)
ROA		1.63	1.62
		(0.03)	(0.03)
Leverage		-0.99	-0.99
		(0.00)	(0.00)
Dividend		0.01	0.01
		(0.11)	(0.12)
Capital Expenditures		-4.60	-4.61
		(0.06)	(0.06)
R&D		3.91	3.90
		(0.00)	(0.00)
Sales/Total Assets		0.16	0.16
		(0.00)	(0.00)
Sales Growth		-0.00	-0.00
		(0.26)	(0.26)
Stock Returns		0.25	0.25
		(0.00)	(0.00)
Loss		1.15	1.15
		(0.01)	(0.01)
Z-Score		-0.01	-0.01
		(0.03)	(0.03)

	(1) Tobin's Q	(2) Tobin's Q	(3) Tobin's Q
E-Index		-0.02	-0.02
		(0.03)	(0.03)
Ln(#News Articles)		-0.01	-0.01
		(0.61)	(0.58)
Inverse Analyst Coverage		0.01	0.01
		(0.88)	(0.89)
Analyst Dispersion		0.00	0.00
		(0.10)	(0.09)
Firm Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
# Obs.	23,030	23,030	23,030
Adj. R^2	0.622	0.706	0.706

Note: *p*-values in parentheses.

Table 4: CEO Twitter Activities and Firm Performance: Moderating Effects

Moderators	(1) Inverse Analyst Coverage	(2) Analyst Dispersion	(3) Ln (#Likes Non-Work-Related)	(4) Ln (#Retweets Non-Work-Related)	(5) %Long-Term
Ln(#Work-Related Tweets)	0.09	0.07	0.10	0.11	0.11
	(0.01)	(0.21)	(0.02)	(0.01)	(0.00)
Ln(#Non-Work-Related Tweets)	-0.09	-0.09	-0.09	-0.02	-0.06
	(0.02)	(0.16)	(0.08)	(0.77)	(0.21)
*Ln(#Work-Related Tweets) * Moderator*	0.07	0.03			
	(0.15)	(0.01)			
*Ln(#Non-Work-Related Tweets) * Moderator*			-0.09	-0.02	-0.06
			(0.00)	(0.00)	(0.08)
Moderator	0.00	0.00	-0.01	0.06	0.03
	(0.97)	(0.01)	(0.51)	(0.07)	(0.25)
Controls	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
# Obs.	23,030	23,030	23,030	23,030	23,030
Adj. R^2	0.706	0.706	0.706	0.706	0.706

Note: *p*-values in parentheses.

Table 5: Event Study on CEO Twitter Adoptions

	Full Sample			Excluding Other News		
	CAR[0,1]	CAR[0,2]	CAR[0,3]	CAR[0,1]	CAR[0,2]	CAR[0,3]
Twitter Adoption	0.24%	0.33%	0.39%	0.22%	0.37%	0.42%
p-Value	0.33	0.33	0.32	0.41	0.39	0.39
#CEO Twitter Adoptions	248	248	248	194	194	194